

ANDERSON SERANGOON JUNIOR COLLEGE

H2 Chemistry 9729

2019 JC2 Prelim Exam Paper 4 Solutions

1 (a) Results

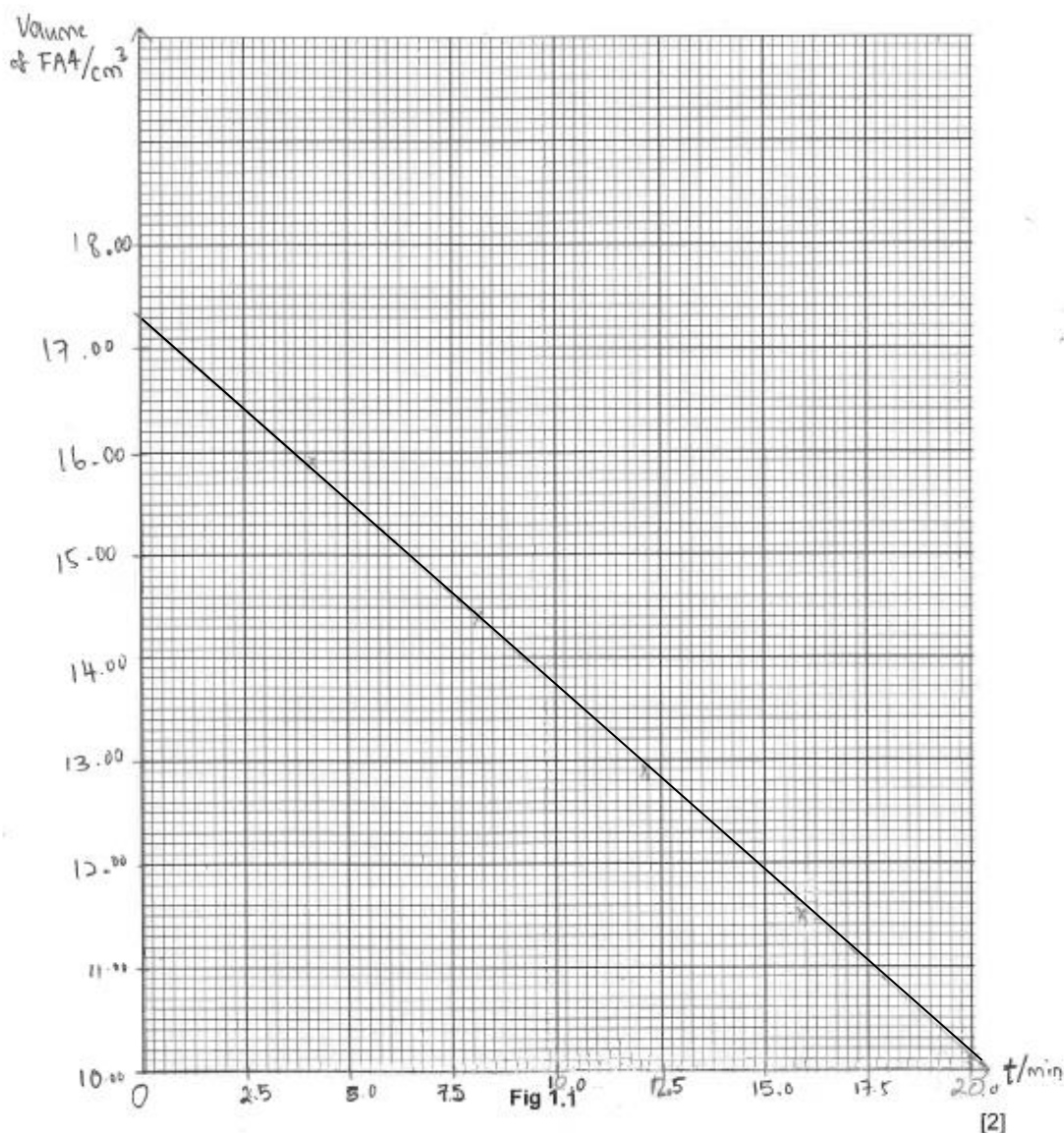
Chosen time/min	Actual time	t/min	Initial burette reading/cm ³	Final burette reading/cm ³	Volume of FA 4 used/cm ³
4	4 min 6s	4.1	0.00	15.90	15.90
8	8 min 3s	8.1	15.90	30.30	14.40
12	12 min 3s	12.1	0.00	12.90	12.90
16	15 min 56s	15.9	12.90	24.40	11.50
20	19 min 59s	20.0	24.40	34.50	10.10

[1]: Tabulates initial and final burette readings, volume of FA 4 added, chosen time, actual time and decimal time with correct headers and units

[1]: record all volumes to 0.05 cm³, actual time in min and s, correct calculation of t to 1 dp

[1]: 5 sets of titration results and each aliquot is taken within ± 0.5 min from the chosen time

(b) (i)



[1]: Both axes are correctly labelled + Scale must be chosen so that plotted points occupy at least half the graph grid in both x and y directions (taking into consideration the extrapolation) + no odd scale

[1]: One plotted point is within $\pm\frac{1}{2}$ small square

- (ii) Order of reaction is zero with respect to I_2 because it is straight line with constant gradient / rate of reaction is independent of $[I_2]$ / $[I_2]$ decreases linearly with time / $[I_2]$ does not affect rate / $[I_2]$ decreases at a constant rate. **[1]**

- (c) (i) Rate = $k [CH_3COCH_3][H^+]$ **[1]**

- (ii) Gradient = $\frac{y_2 - y_1}{x_2 - x_1} = \frac{9 - 14.8}{18.75} = \underline{-0.309 \text{ cm}^3 \text{ min}^{-1}}$ (3 s.f with units) **[1]**

Show coordinates on the graph or in the working

- (iii) rate of change = (c)(ii) $\times 0.01/1000$
 $= 0.309 \times 0.01/1000$
 $= 3.09 \times 10^{-6}$
 $= \underline{3.09 \times 10^{-6} \text{ mol min}^{-1}}$ (3 s.f.) **[1]**

- (iv) $I_2 \equiv 2S_2O_3^{2-}$
rate of disappearance of $I_2 = \frac{1}{2} \times \text{(c)(iii)}$
 $= \frac{1}{2} \times 3.09 \times 10^{-6}$
 $= 1.545 \times 10^{-6}$
 $= \underline{1.55 \times 10^{-6} \text{ mol min}^{-1}}$ (3 s.f.) **[1]**

- (v) rate of change of $[I_2] = \text{(c)(iv)} \div 10/1000$
 $= 1.545 \times 10^{-6} \div 10/1000$
 $\approx \underline{1.55 \times 10^{-4} \text{ mol dm}^{-3} \text{ min}^{-1}}$ (3 s.f with units) **[1]**

- (vi) (c)(v) = $k[CH_3COCH_3][H^+]$
 $1.545 \times 10^{-4} = k [CH_3COCH_3][H^+]$
 $1.545 \times 10^{-4} = k (1.00 \times \frac{1}{4})(1.00 \times 2 \times \frac{1}{4})$
 $k = \underline{1.24 \times 10^{-3} \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}}$ (3 s.f with units) **[1]**

- (d) It is added to react with acid catalyst so that the reaction will stop / It is to quench the reaction by reacting with the acid catalyst. **[1]**

- (e) The low concentration of iodine means that very little propanone and acid are reacted away from the reaction mixture and hence the concentration of propanone and acid remain effectively constant. Hence, the order of reaction with respect to iodine can be determined because any change in the rate is due to the change in concentration of iodine. **[1]**

- (f) (i) nucleophilic substitution **[1]**

- (ii) To prevent reaction of $CH_3CH_2CH_2Br$ with water/hydrolysis (if wet) **[1]**

- (iii) Due to the low solubility of NaBr in propanone, NaBr formed gets precipitated out. Position of equilibrium shifts to the right. **[1] or**

Since sodium bromide is not soluble in water, there will not be sufficient Br^- ions to attack the electrophilic C attached to I, hence position of equilibrium favors right **or**

Since NaI is much more soluble than NaBr, $[NaI]$ is $\gg [NaBr]$. Forward rate is much greater than the backward rate. Hence, the reaction favors right.

- (iv) Since 100% yield of pure 1-iodopropane is obtained and sodium iodide is used in excess,

$$n(\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}) \text{ required} = n(\text{CH}_3\text{CH}_2\text{CH}_2\text{I}) \text{ obtained} \\ = 10 / 169.9 = 0.05886 \text{ mol}$$

$$\text{mass of CH}_3\text{CH}_2\text{CH}_2\text{Br} = 0.05886 \times 122.9 = 7.234 \text{ g}$$

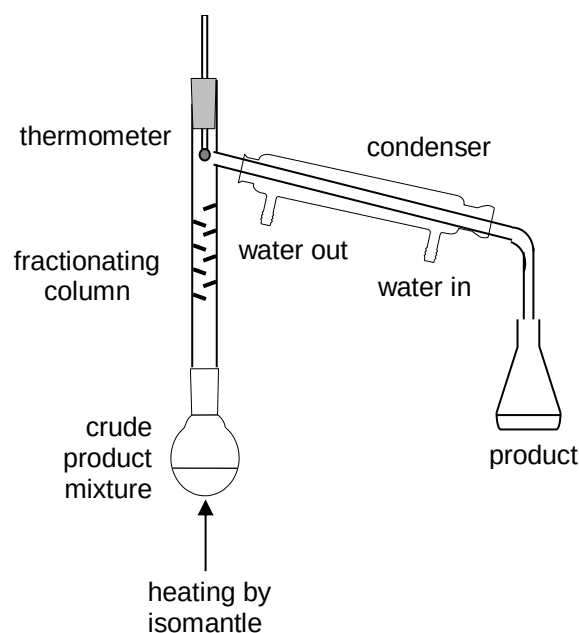
$$\text{minimum volume of CH}_3\text{CH}_2\text{CH}_2\text{Br} = 7.234 / 1.35 = \underline{5.36 \text{ cm}^3} \quad [1]$$

- (v) Minimum mass of sodium iodide required = $0.05885 \times 149.9 = 8.82 \text{ g}$
Since sodium iodide is added in excess, mass of sodium iodide to be used $> 8.82 \text{ g}$

Preparation of crude 1-iodopropane

1. Weigh accurately about 10 g of sodium iodide in a 50 cm³ round-bottomed flask.
2. Using a 10 cm³ measuring cylinder, transfer 6 cm³ of 1-bromopropane to the flask.
3. Using a 50 cm³ measuring cylinder, transfer 25 cm³ of propanone to the flask.
4. Swirl the flask to ensure even mixing.
5. Add boiling chips (anti-bumping granules) to the mixture.
6. Using a water bath / isomantle, gentle heat the flask fitted with a reflux condenser for about 30 minutes.
7. Cool down the flask by removing the water bath / isomantle.
8. Remove the flask and fractionally distil the mixture to obtain the pure 1-iodopropane at its boiling point of 102.6 °C.

(Note: solid NaI and NaBr will remain in the flask)



max. [2] for labelled diagram (fractionating column, thermometer, condenser, heating by isomantle clearly labelled)

[1]: Sound quantities of 1-bromopropane, sodium iodide and propanone chosen for preparation of crude 1-iodopropane

[1]: General procedure for preparation of crude 1-iodopropane

[1]: Apparatus used for measurements (measuring cylinders) & reflux (round-bottomed flask, reflux condenser, hot water bath / isomantle)

2 (a) (i) Results

mass of empty weighing bottle / g	
mass of weighing bottle + KI / g	
mass of weighing bottle + residual KI / g	
mass of KI used / g	2.013

initial temperature of water / °C	
minimum temperature reached / °C	
ΔT_{KI} / °C	2.0

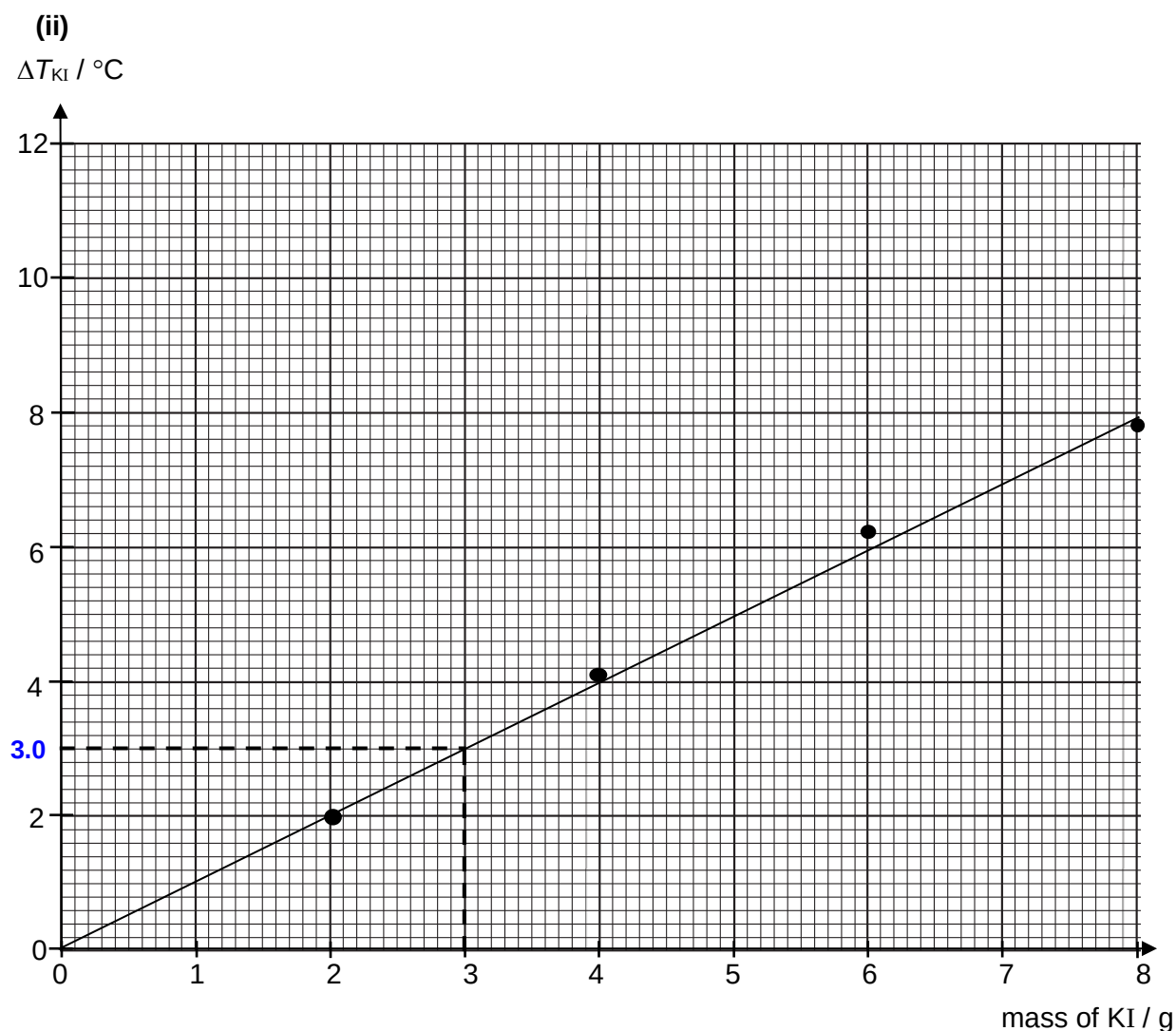
[1]: Both tables have correct headers and units

[1]: Weighs between 1.950 g to 2.050 g of KI **and** correctly calculates mass of KI used and correctly calculates ΔT_{KI}

Final mass of KI dissolved may be out of range, but initially weighed mass must be within range.

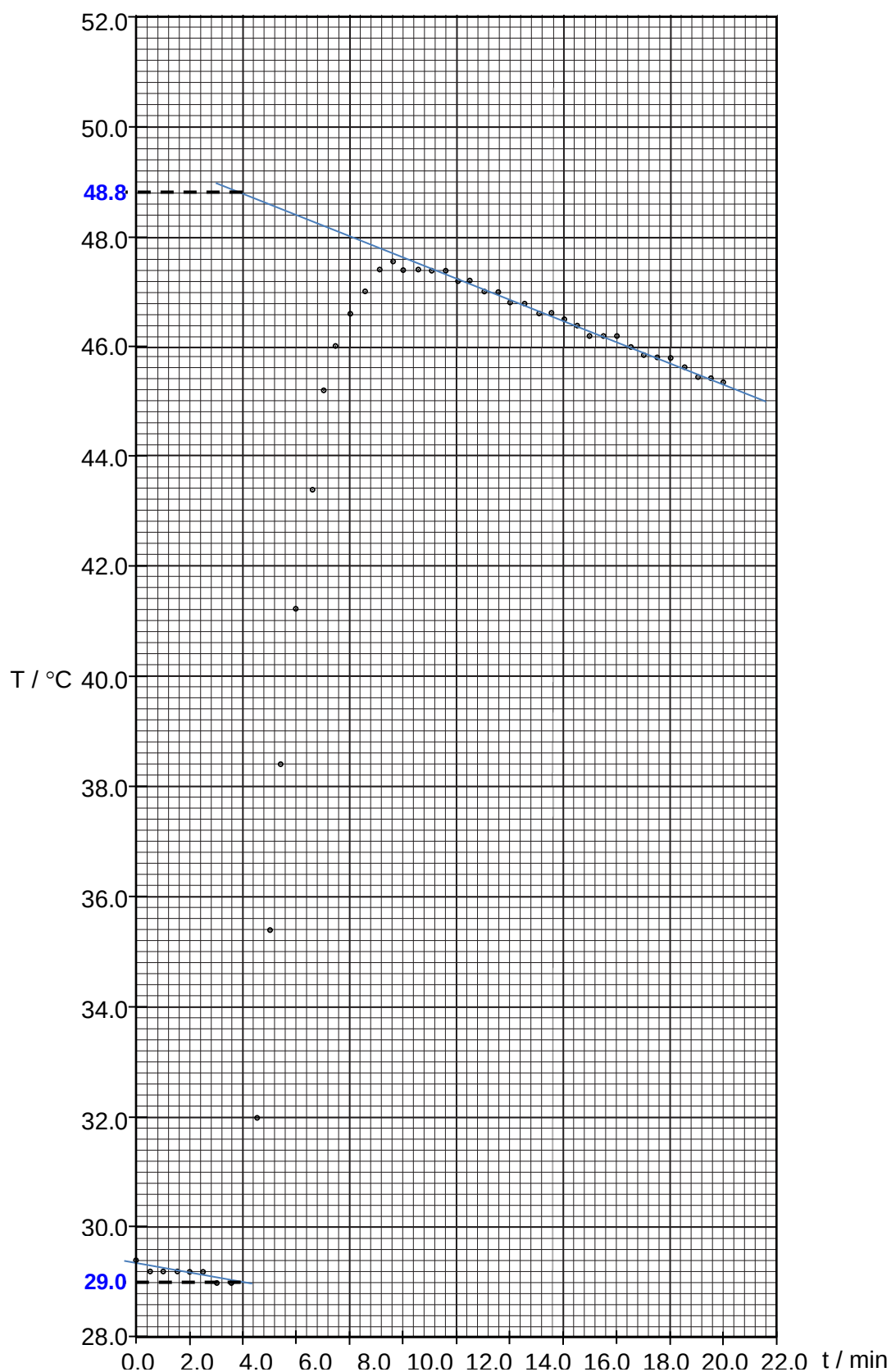
[1]: All temperature recorded and calculated ΔT_{KI} value are to 0.1 °C, **and** all mass recorded calculated mass of KI used are to 3 d.p.

[1]: Accuracy: how close the point is to the teachers' result (2 small square from teachers result)



[1]: Correctly plots all points and draws line of best-fit which passes through the origin

(b)



$\Delta T_m = 48.8 - 29.0 = \underline{19.8} \text{ } ^\circ\text{C}$ (read T_{max} and T_{min} correctly to $\pm \frac{1}{2}$ small square, i.e. to $0.1 \text{ } ^\circ\text{C}$ (1 d.p))

[1]: Draws best-fit line (instead of horizontal line) for the points before FA 6 was added.

[1]: Correctly calculates ΔT_m (allow $\pm 0.4 \text{ } ^\circ\text{C}$ difference from teachers' value because of 1 small square ($\pm 0.2 \text{ } ^\circ\text{C}$) allowance for each temperature reading).

- (c) (i) **[1]:** Correctly reads off ΔT_{KI} for m g of potassium iodide from plotted graph at the point

$$m = \frac{\text{mass of FA 6 used}}{2} = \frac{6.020}{2} = 3.010 \text{ g}$$

*Or correctly calculates using ΔT per gram of KI by finding gradient
Working on the graph, e.g. dotted lines or otherwise, must be shown.*

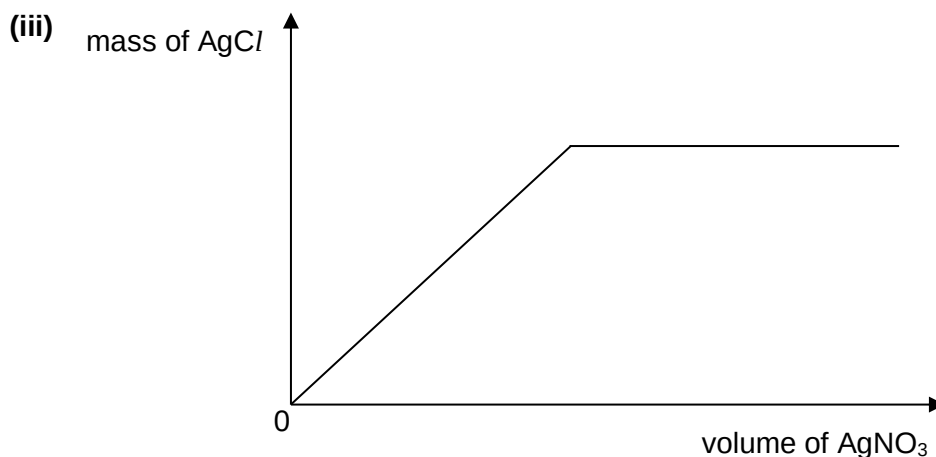
(ΔT_{KI} : read to $\pm \frac{1}{2}$ small square, i.e. to 0.1°C (1 d.p))

- (ii) **[1]:** Correctly calculates change in temperature due to lithium chloride,
 $\Delta T_m - \Delta T_{KI} = \Delta T_{LiCl}$ (to 0.1°C (1 d.p))
- (iii) **[1]:** Correctly calculates the value of change in temperature of water per gram of lithium chloride = $\frac{\Delta T_{LiCl}}{(6.020 \div 2)}$ (3 s.f)

[1]: Correct significant figures for 1(c)(ii) to (c)(vi); 2(b), (c)(i) to (c)(iii)

[1]: Correct units for 1(c)(ii),(v),(vi)

- (d) (i) Insoluble barium chromate is formed. **[1]**
- (ii) The formation of insoluble barium sulfate will ensure that insoluble barium chromate is not formed during titration. **[1]**



*Straight line through origin **and** reaches a plateau **[1]***

- (iv) 1. Weigh accurately about 3.00 g of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ in a pre-weighed weighing bottle, using a weighing balance (or electronic balance)
2. Dissolve this solid in a beaker with 30 cm^3 of water.
3. Transfer the solution and washings into a 250 cm^3 graduated flask and make up to the mark with deionised water
4. Shake the solution to obtain a homogeneous solution.
5. Reweigh the emptied weighing bottle.

[1]: Sound procedure for standard solution preparation (weigh solid, dissolve in water, transfer, make up to the mark)

[1]: Ensures accuracy and reliability for preparation of standard solution (use of graduated flask, transfer washings, shake, reweigh if solid not completely transferred to graduated flask)

- (v) First: sulfuric acid
Second: potassium chromate(VI)
Third: silver nitrate

3 (a)

	test	observations
1.	Add 1 cm depth of FA 7 to a test-tube. Add aqueous ammonia slowly, with shaking, until no further change is seen. Filter the mixture and add dilute nitric acid drop-wise to the filtrate, until no further change is seen. (You may continue with test 2, while waiting for the filtration process to complete.)	(Off) white ppt, insoluble in excess NH_3 Filtrate is colourless Residue is brown in colour White ppt reappears, soluble in excess acid
2.	Add 1 cm depth of FA 7 to a test-tube. Add aqueous barium nitrate to FA 7, followed by dilute nitric acid.	White ppt formed White ppt insoluble in acid

[3]

FA 7 contains the cations Mn^{2+} and Zn^{2+} .

identity	evidence
FA 7 contains cation: Mn^{2+}	Off white ppt, insoluble in excess NH_3 Brown ppt formed on standing
FA 7 contains cation: Zn^{2+}	White ppt soluble in excess NH_3 White ppt reforms when dilute acid is added dropwise

FA 7 contains the anion SO_4^{2-} .

[2]

(b) (i)

	test	observations
1.	<i>Add 1 cm depth of FA 8 to a test-tube.</i> <i>Add FA 4 slowly, with shaking, until no further change is seen.</i> <i>Leave it to stand.</i>	<u>Yellow solution turns brown/reddish brown/dark orange/violet</u> <u>Yellow solution reforms</u>
2.	<i>Add 1 cm depth of FA 8 to a test-tube.</i> <i>Add 1 cm depth of FA 5, with shaking, until no further change is seen.</i>	Orange/red-brown ppt. <u>Effervescence</u> observed. Colourless gas evolved gives a <u>white ppt</u> with Ca(OH)_2 / limewater.
3.	<i>Add 1 cm depth of FA 8 to a test-tube.</i> <i>Add 1 cm depth of aqueous potassium iodide, with shaking, until no further change is seen.</i>	<u>Yellow solution turns brown.</u>

[2]

- (ii) Fe^{3+} has a high charge density. Hence, the cation can polarise the water molecules, giving rise to an acidic solution.



The acidic solution undergoes acid-base reaction with NaHCO_3 to give CO_2 and a red-brown ppt of $\text{Fe}(\text{OH})_3$.

- (iii) • Fe^{3+} is an oxidising agent. Fe^{3+} oxidises I^- to brown I_2 [1]

- Reaction between $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}(\text{aq})$ and I^- :
 $E^\circ_{\text{cell}} = +0.77 - 0.54 = +0.23 \text{ V} > 0 \Rightarrow$ reaction is feasible [1]

- (iv) Reaction between $[\text{Fe}(\text{CN})_6]^{3-}$ and I^- :
 $E^\circ_{\text{cell}} = +0.36 - 0.54 = -0.18 \text{ V} < 0 \Rightarrow$ reaction is not feasible.

[1]: Correct calculation of E°_{cell} (E° values quoted)

CN^- ligands stabilise Fe^{3+} more than Fe^{2+} , thus $E^\circ([\text{Fe}(\text{CN})_6]^{3-}/[\text{Fe}(\text{CN})_6]^{4-})$ is less positive than $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+})$. $[\text{Fe}(\text{CN})_6]^{3-}$ is less easily reduced / is a weaker oxidising agent than $\text{Fe}^{3+}(\text{aq})$. [1]

- (v) • Purple organic layer
• Yellow / Brown aqueous layer [1]

Note: Both layers are immiscible.